Web Application Development Lab 05 Report

Student Name: Vũ Quốc Khánh
Student ID: ITCSIU24039

Exercise 1

Task 1.2

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package model;


import java.sql.Timestamp;


/**
 *
 * @author Islabre
 */
public class Student {
    private int id;
    private String studentCode;
    private String fullName;
    private String email;
```

```
    private String major;
    private Timestamp createdAt;

    public Student() {
    }

    public Student(String studentCode, String fullName, String email,
String major) {
        this.studentCode = studentCode;
        this.fullName = fullName;
        this.email = email;
        this.major = major;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getStudentCode() {
        return studentCode;
    }

    public void setStudentCode(String studentCode) {
```

```
        this.studentCode = studentCode;
    }

    public String getFullName() {
        return fullName;
    }

    public void setFullName(String fullName) {
        this.fullName = fullName;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public String getMajor() {
        return major;
    }

    public void setMajor(String major) {
        this.major = major;
    }
```

```
    public Timestamp getCreatedAt() {
        return createdAt;
    }

    public void setCreatedAt(Timestamp createdAt) {
        this.createdAt = createdAt;
    }

    @Override
    public String toString() {
        return "Student{" + "id=" + id + ", studentCode=" +
studentCode + ", fullName=" + fullName + ", email=" + email + ",
major=" + major + '}';
    }
}
```

Task 1.3

Code

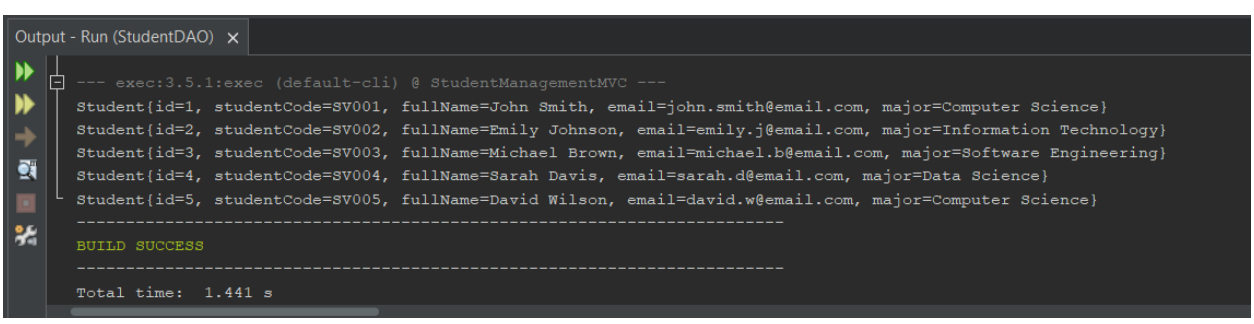
```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java
to edit this template
 */
package dao;

import model.Student;
import java.sql.*;
```

```
import java.util.ArrayList;
import java.util.List;

/**
 *
 * @author Islabre
 */
public class StudentDAO {
    private final String url =
"jdbc:mysql://localhost:3306/student_management";
    private final String username = "root";
    private final String password = "Q@jaP*96T8xY#2^7&3@7";

    // Add this main method to test (remove after testing)
    public static void main(String[] args) {
        StudentDAO dao = new StudentDAO();
        List<Student> students = dao.getAllStudents();
        for (Student s : students) {
            System.out.println(s);
        }
    }

    private Connection getConnection() throws SQLException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            return DriverManager.getConnection(this.url,
this.username, this.password);
```

```
        } catch (ClassNotFoundException e) {
            throw new Error("Error: MySQL Driver not found");
        }
    }

    public List<Student> getAllStudents() {
        Connection conn = null;
        PreparedStatement pstmt = null;
        ResultSet rs = null;

        List<Student> students = new ArrayList<>();

        String sql = "SELECT * FROM students ORDER BY id";

        try {
            conn = getConnection();
            pstmt = conn.prepareStatement(sql);
            rs = pstmt.executeQuery();

            while (rs.next()) {
                Student s = new Student(
                    rs.getString("student_code"),
                    rs.getString("full_name"),
                    rs.getString("email"),
                    rs.getString("major")
                );
```

```
                s.setId(rs.getInt("id"));
                s.setCreatedAt(rs.getTimestamp("created_at"));

                students.add(s);
            }

            return students;
        } catch (SQLException e) {
            throw new Error("Database Error: " + e.getMessage());
        } finally {
            try {
                if (conn != null) conn.close();
                if (pstmt != null) pstmt.close();
                if (rs != null) rs.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Testing Result



Exercise 2

Task 2.1

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java
to edit this template
 */
package controller;


import dao.StudentDAO;
import model.Student;


import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.util.List;


/**
 *
 * @author Islabre
```

```
 */
@WebServlet("/student")
public class StudentController extends HttpServlet {
    private StudentDAO studentDAO;

    @Override
    public void init() {
        this.studentDAO = new StudentDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String action = request.getParameter("action");

        if (action == null) action = "list";

        // TODO: Route to appropriate method based on action
        switch (action) {
            case "list":
                listStudents(request, response);
                break;
            // TODO: Add more cases later
        }
    }
```

```
    private void listStudents(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        List<Student> students = studentDAO.getAllStudents();
        request.setAttribute("students", students);


        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-list.jsp");
        dispatcher.forward(request, response);
    }
}
```

Task 2.2

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java
to edit this template
 */
package controller;


import dao.StudentDAO;
import model.Student;


import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.util.List;

/**
 *
 * @author Islabre
 */
@WebServlet("/student")
public class StudentController extends HttpServlet {
    private StudentDAO studentDAO;

    @Override
    public void init() {
        this.studentDAO = new StudentDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String action = request.getParameter("action");

        if (action == null) action = "list";

        // TODO: Route to appropriate method based on action
```

```
        switch (action) {
            case "list": listStudents(request, response); break;
            case "new": showNewForm(request, response); break;
            case "edit": showEditForm(request, response); break;
            case "delete": deleteStudent(request, response); break;
        }
    }

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response) throws ServletException, IOException {
       String action = request.getParameter("action");

        switch (action) {
            case "insert": insertStudent(request, response); break;
            case "update": updateStudent(request, response); break;
        }
    }



    private void listStudents(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        List<Student> students = studentDAO.getAllStudents();
        request.setAttribute("students", students);

        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-list.jsp");
```

```
        dispatcher.forward(request, response);
    }

    private void showNewForm(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-form.jsp");
        dispatcher.forward(request, response);
    }

    private void showEditForm(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String id = request.getParameter("id");
        // TODO: Get student by id and forward as attribute

        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-form.jsp");
        dispatcher.forward(request, response);
    }

    private void deleteStudent(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String id = request.getParameter("id");
        // TODO: Delete student by id
```

```
        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-list.jsp");

        dispatcher.forward(request, response);

    }


    private void insertStudent(HttpServletRequest request,
HttpServletResponse response)

            throws ServletException, IOException {

        // TODO: Insert student from POST info


        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-list.jsp");

        dispatcher.forward(request, response);

    }


    private void updateStudent(HttpServletRequest request,
HttpServletResponse response)

            throws ServletException, IOException {

        // TODO: Update student from POST info


        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-list.jsp");

        dispatcher.forward(request, response);

    }

}
```

Exercise 3

Task 3.1

Code

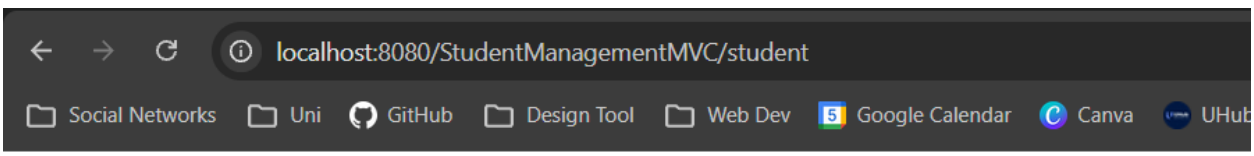
```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<!DOCTYPE html>

<html>

<head>

    <meta charset="UTF-8">

    <title>Student List</title>

    <style>

        table, td, th {

            border: 1px solid;

            border-collapse: collapse;

        }


        th, td {

            padding: 0.25em;

        }


        button a {

            color: inherit;

            text-decoration: none;

        }

    </style>
```

```
</head>
<body>
    <h1>Student Management (MVC)</h1>

    <c:if test="${not empty param.message}">
        <div class="success">${param.message}</div>
    </c:if>

    <c:if test="${not empty param.error}">
        <div class="success">${param.error}</div>
    </c:if>

    <button>
        <a href="student?action=new">
            Add Student
        </a>
    </button>

    <table>
        <thead>
            <tr>
                <th>ID</th>
                <th>Code</th>
                <th>Name</th>
                <th>Email</th>
                <th>Major</th>
                <th>Actions</th>
```

```
            </tr>
        </thead>
        <tbody>
            <c:forEach var="student" items="${students}">
                <tr>
                    <td>${student.id}</td>
                    <td>${student.studentCode}</td>
                    <td>${student.fullName}</td>
                    <td>${student.email}</td>
                    <td>${student.major}</td>
                    <td>
                        <button>
                            <a
href="student?action=edit&id=${student.id}">
                                Edit
                            </a>
                        </button>

                        <button>
                            <a
href="student?action=delete&id=${student.id}">
                                Delete
                            </a>
                        </button>
                    </td>
                </tr>
            </c:forEach>
```

```
            <c:if test="${empty students}">
                <tr>
                    <td colspan="6">No students found</td>
                </tr>
            </c:if>
        </tbody>
    </table>
</body>
</html>
```

Result



Student Management (MVC)

Add Student

ID	Code	Name	Email	Major	Actions
1	SV001	John Smith	john.smith@email.com	Computer Science	Edit Delete
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	Edit Delete
3	SV003	Michael Brown	michael.b@email.com	Software Engineering	Edit Delete
4	SV004	Sarah Davis	sarah.d@email.com	Data Science	Edit Delete
5	SV005	David Wilson	david.w@email.com	Computer Science	Edit Delete

Task 3.2

Code

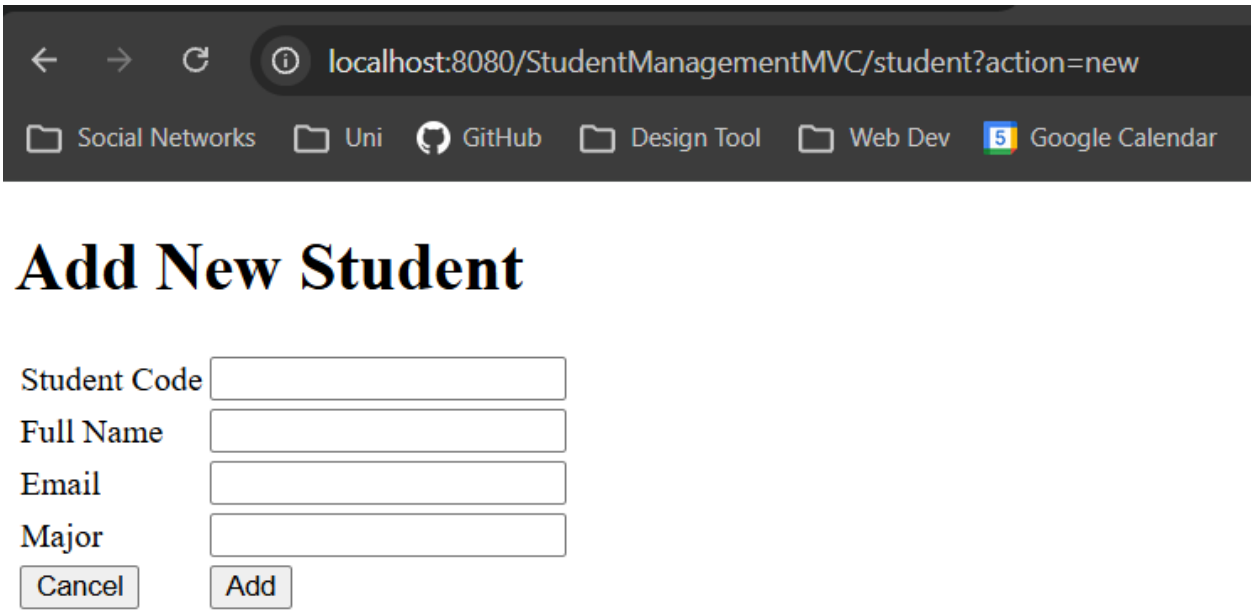
```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<!DOCTYPE html>

<html>

<head>

    <title>

        <c:choose>

            <c:when test="${student != null}">Edit Student</c:when>

            <c:otherwise>Add New Student</c:otherwise>

        </c:choose>

    </title>

    <style>

        button a {

            color: inherit;

            text-decoration: none;

        }


        input[readonly] {

            cursor: not-allowed;

        }

    </style>

</head>

<body>

    <div class="container">
```

```
        <h1>
            <c:choose>
                <c:when test="${student != null}">Edit
Student</c:when>
                <c:otherwise>Add New Student</c:otherwise>
            </c:choose>
        </h1>


        <form action="student" method="POST">
            <input type="hidden" name="action" value="${student !=
null ? 'update' : 'insert'}">


            <c:if test="${student != null}">
                <input type="hidden" name="id"
value="${student.getId()}">
            </c:if>


            <table>
                <tr>
                    <td>Student Code</td>
                    <td>
                        <input type="text" name="studentCode"
                               value="${student.getStudentCode()}"
${student != null ? 'readonly' : 'required'}>
                    </td>
                </tr>
                <tr>
                    <td>Full Name</td>
```

```
                    <td>
                        <input type="text" name="fullName"
value="${student != null ? student.getFullName() : ''}">
                    </td>
                </tr>
                <tr>
                    <td>Email</td>
                    <td>
                        <input type="email" name="email"
value="${student != null ? student.getEmail() : ''}">
                    </td>
                </tr>
                <tr>
                    <td>Major</td>
                    <td>
                        <input type="text" name="major"
value="${student != null ? student.getMajor() : ''}">
                    </td>
                </tr>
                <tr>
                    <td>
                        <button>
                            <a href="student?action=list">Cancel</a>
                        </button>
                    </td>
                    <td>
                        <input type="submit" value="${student != null
? 'Save' : 'Add'}">
```

```
                    </td>
                </tr>
            </table>
        </form>
    </div>
</body>
</html>
```

Result



Exercise 4

Task 4.1

```
/*
```

```
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java
to edit this template
 */
package controller;

import dao.StudentDAO;
import model.Student;

import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.util.List;

/**
 *
 * @author Islabre
 */
@WebServlet("/student")
public class StudentController extends HttpServlet {
    private StudentDAO studentDAO;
```

```
    @Override
    public void init() {
        this.studentDAO = new StudentDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String action = request.getParameter("action");

        if (action == null) action = "list";

        // TODO: Route to appropriate method based on action
        switch (action) {
            case "list": listStudents(request, response); break;
            case "new": showNewForm(request, response); break;
            case "edit": showEditForm(request, response); break;
            case "delete": deleteStudent(request, response); break;
        }
    }

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response) throws ServletException, IOException {
       String action = request.getParameter("action");
```

```
        switch (action) {
            case "insert": insertStudent(request, response); break;
            case "update": updateStudent(request, response); break;
        }
    }



    private void listStudents(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        List<Student> students = studentDAO.getAllStudents();
        request.setAttribute("students", students);


        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-list.jsp");
        dispatcher.forward(request, response);
    }


    private void showNewForm(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-form.jsp");
        dispatcher.forward(request, response);
    }


    private void showEditForm(HttpServletRequest request,
HttpServletResponse response)
```

```
            throws ServletException, IOException {
        String id = request.getParameter("id");
        Student student =
studentDAO.getStudentById(Integer.valueOf(id));

        if (student != null) {
            request.setAttribute("student", student);
        }

        RequestDispatcher dispatcher =
request.getRequestDispatcher("/views/student-form.jsp");
        dispatcher.forward(request, response);
    }

    private void deleteStudent(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String id = request.getParameter("id");

        boolean success =
studentDAO.deleteStudent(Integer.valueOf(id));
        String message = "";

        if (success) {
            message = "Student deleted successfully";
            response.sendRedirect("student?action=list&message=" +
message);
        } else {
```

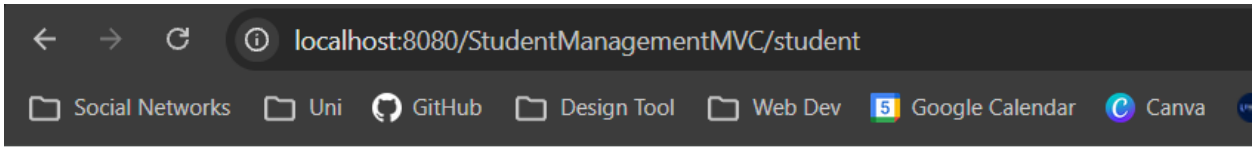
```
            message = "Error: Failed to delete student";
            response.sendRedirect("student?action=list&error=" +
message);
        }
    }

    private void insertStudent(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String studentCode = request.getParameter("studentCode");
        String fullName = request.getParameter("fullName");
        String email = request.getParameter("email");
        String major = request.getParameter("major");

        Student student = new Student(studentCode, fullName, email,
major);

        boolean success = studentDAO.addStudent(student);
        String message = "";

        if (success) {
            message = "Student added successfully";
            response.sendRedirect("student?action=list&message=" +
message);
        } else {
            message = "Error: Failed to add student";
            response.sendRedirect("student?action=list&error=" +
message);
```

```
        }
    }

    private void updateStudent(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        String id = request.getParameter("id");
        String studentCode = request.getParameter("studentCode");
        String fullName = request.getParameter("fullName");
        String email = request.getParameter("email");
        String major = request.getParameter("major");

        Student student = new Student(studentCode, fullName, email,
major);
        student.setId(Integer.valueOf(id));

        boolean success = studentDAO.updateStudent(student);
        String message = "";

        if (success) {
            message = "Student updated successfully";
            response.sendRedirect("student?action=list&message=" +
message);
        } else {
            message = "Error: Failed to update student";
            response.sendRedirect("student?action=list&error=" +
message);
        }
```

```
    }
}
```

Task 4.2

1. List:



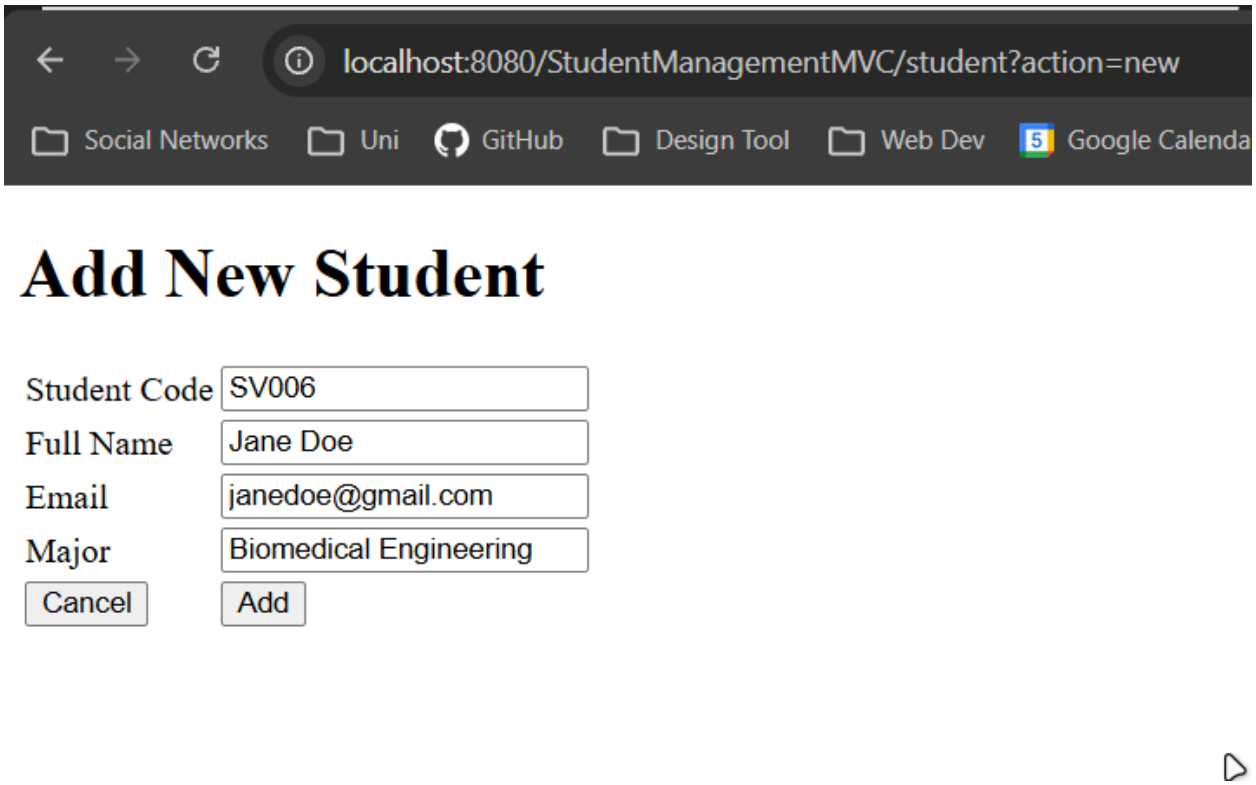
Student Management (MVC)

Add Student

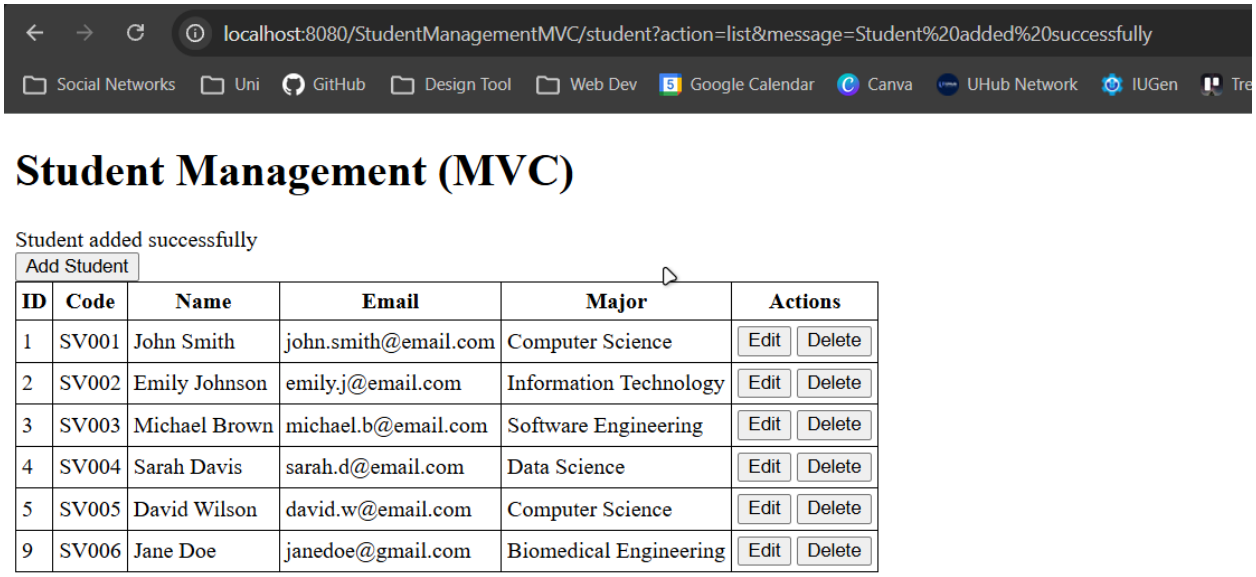
ID	Code	Name	Email	Major	Actions
1	SV001	John Smith	john.smith@email.com	Computer Science	Edit Delete
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	Edit Delete
3	SV003	Michael Brown	michael.b@email.com	Software Engineering	Edit Delete
4	SV004	Sarah Davis	sarah.d@email.com	Data Science	Edit Delete
5	SV005	David Wilson	david.w@email.com	Computer Science	Edit Delete

2. Add:

- Form with sample data:

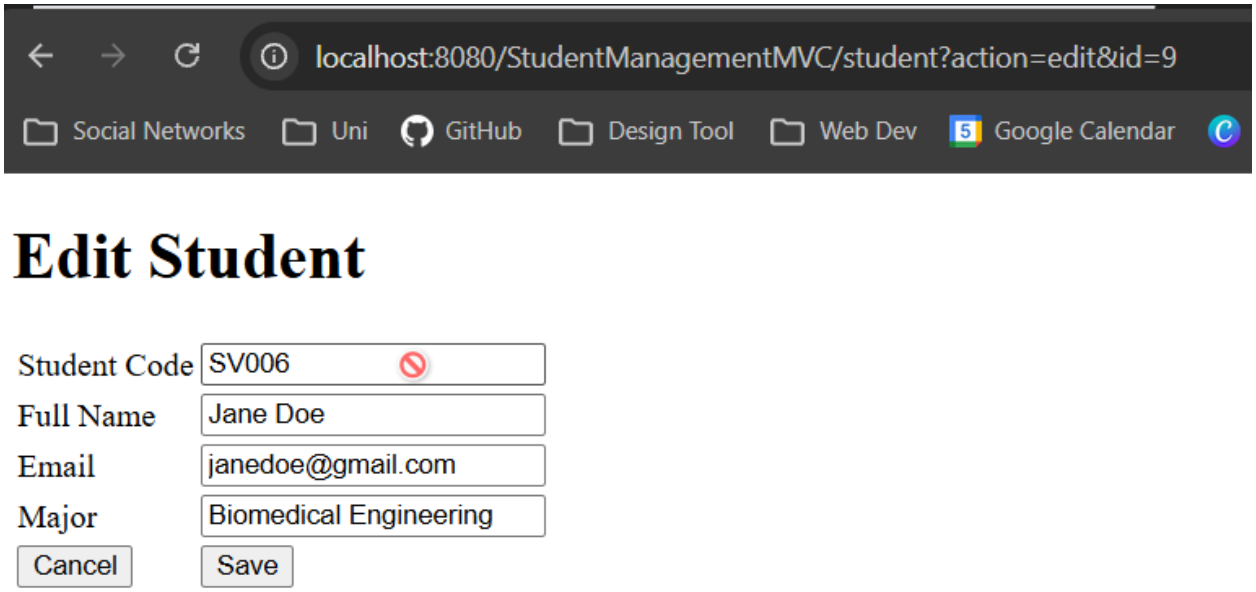


- After submitting:

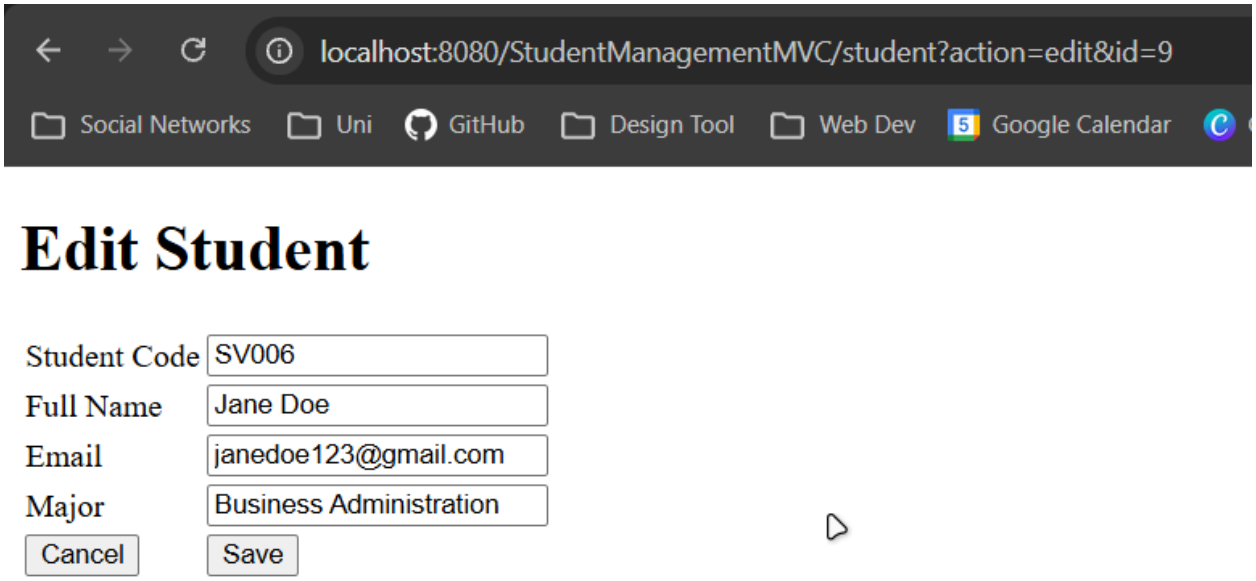


3. Edit:

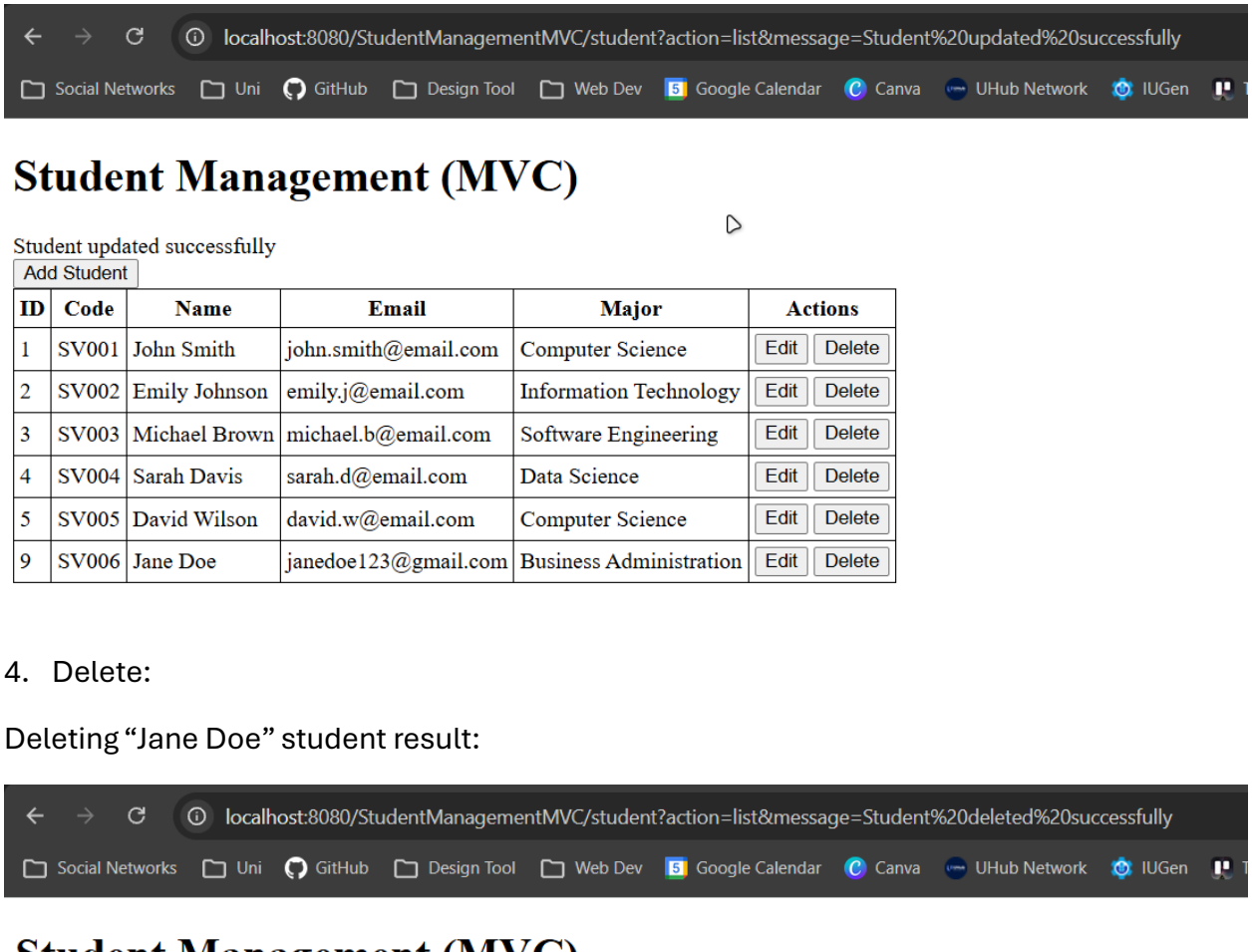
- Editing “Jane Doe” student:

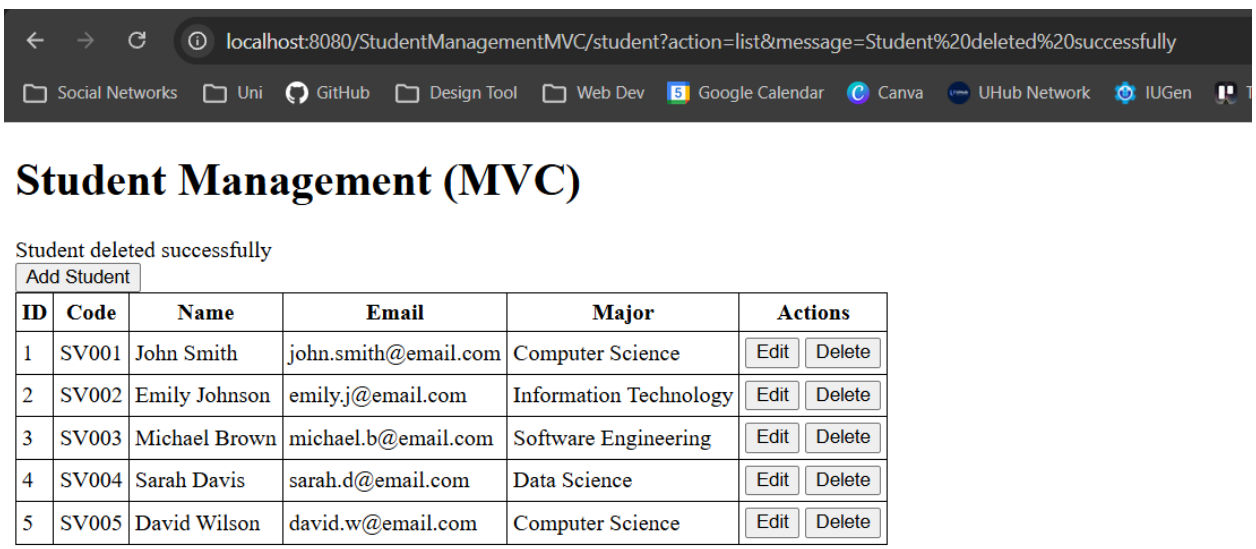


- Modified student data:



- After submitting:



4. Delete:

Deleting “Jane Doe” student result: